Ravi floodplain, Lahore District: pre-monsoon exposure report

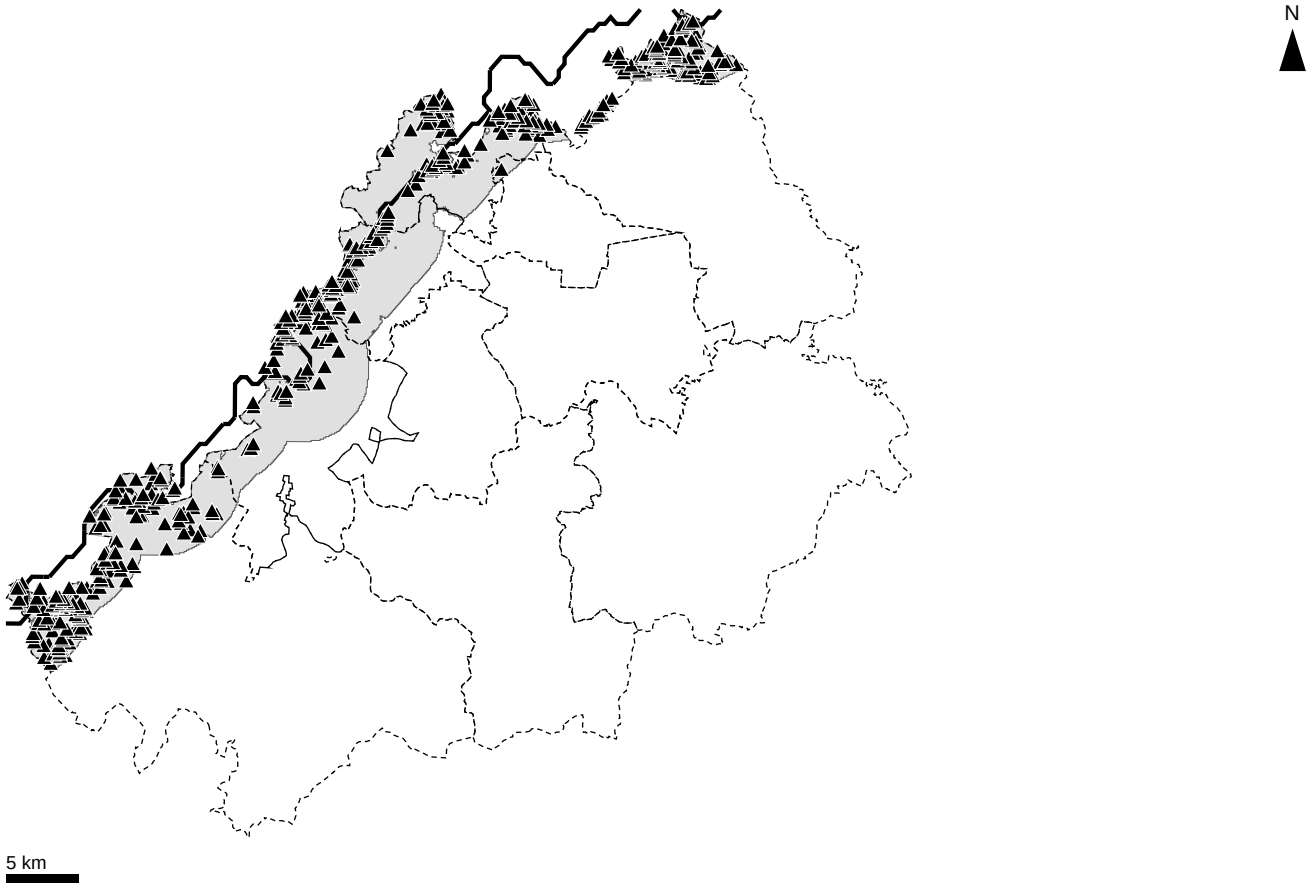
Kinara · Ravi Floodplain Watch · prepared 6 Nov 2026

Floodplain definition: **Floodplain (HAND proxy)** (243.3 km²; land within 4,000 m of the Ravi and under 5 m above the nearest drainage). Not derived from the 2025 flood.

Estimated buildings in the floodplain	**216,258**
First seen after 2016 (2017-2023)	**38,932 (18%)**
In the Aug 2025 flood extent (satellite)	**1,354 (thresholds not yet calibrated)**
Estimated population in the floodplain	**about 1,863,000**
Open candidate alerts (new + visited)	**510**

Buildings: Open Buildings v3 (footprints, imagery about 2022-23) and 2.5D Temporal (2016-2023). Population: WorldPop R2025A, 2025 (100 m). Flood: Sentinel-1 orbit 107, 28 Aug 2025.

Map



Grey: floodplain zone. Dashed: tehsil boundaries. Thick line: Ravi. Triangles: candidate alerts.

Tehsils

Tehsil (part in floodplain)	km²	Buildings	After 2016	Flooded 2025	Population
Ravi	53.95	82,533	14,442	588	666,200
Lahore City	39.7	70,664	6,994	266	735,300
Allama Iqbal	56.5	43,343	9,642	359	395,100
Lahore Raiwand	66.28	11,813	5,319	95	49,000
Wahga	24.9	2,284	718	1	8,400
Lahore Shalamar	1.97	1,677	1,142	20	8,500

Top 20 mouzas by estimated buildings in the floodplain

#	Mouza	Tehsil	Buildings	After 2016	Flooded 2025	Population
1	Shahdara *	Allama Iqbal	96,607	12,675	517	869,000
2	Bela Basti Raam	Ravi	15,033	1,581	172	158,700
3	Kot Mohibbu	Ravi	7,340	787	41	59,000
4	Kot Begum	Ravi	7,116	1,174	41	60,400
5	Mohlan Wal	Lahore Raiwand	6,218	3,580	32	27,400
6	Sanda Khurd	Lahore City	5,866	385	19	60,900
7	Raaj Garh	Lahore City	5,843	83	12	62,200
8	Shah Pur (Khan Pur)	Allama Iqbal	5,257	1,107	83	58,600
9	Chung Panjgrain	Allama Iqbal	4,646	1,016	55	27,600
10	Nain Sukh	Ravi	4,206	1,338	30	35,200
11	Lahore Khaas	Lahore City	3,676	300	26	77,600
12	Babu Sabu	Lahore City	3,645	1,382	28	28,500
13	Kanjrah	Allama Iqbal	3,141	1,922	28	18,300
14	Ganjar Kalan	Ravi	2,673	867	19	16,100
15	Khud Pur	Lahore Raiwand	2,583	821	22	8,100
16	Rakh Sheikh Kot	Allama Iqbal	2,563	738	29	18,300
17	Dholan Waal	Lahore City	2,469	95	3	21,200
18	Bhama	Ravi	2,333	820	14	15,500
19	Chak Jia Musa	Lahore City	2,168	142	11	25,100
20	Khoee Miran	Ravi	2,132	108	7	22,200

* The mouza polygon crosses more than one official tehsil boundary (the two boundary layers disagree); its tehsil is the one it overlaps most. Population rounded to the nearest 100.

Candidate new-construction alerts

510 alerts in total. New: 510

Tehsil	Alerts
Lahore Raiwand	177
Wahga	114
Ravi	97
Allama Iqbal	61
Lahore City	60
Lahore Shalamar	1

Verified precision: no verified alerts yet.

Method and caveats

Buildings. Google Open Buildings v3 footprints (confidence at least 0.75) inside the floodplain. A footprint is a roof seen from space; counts are estimated buildings, not households.

Year first seen. Open Buildings 2.5D Temporal, building presence at least 0.5 per year 2016-2023. 2016 means on or before 2016. About 4% of footprints have no year (not seen by the temporal model).

Flood 2025. Sentinel-1 radar before (Jul to mid-Aug) and during (26 Aug to 5 Sep 2025) the flood, same orbit. Open land: darker than -18 dB and 3 dB darker than before. Built-up: 3 dB brighter (water against walls). **Thresholds are not yet calibrated against field or news locations; the flooded building count is provisional and probably low.**

Alerts. Radar backscatter at least 3 dB above a 90-day baseline on two consecutive passes of the same orbit, inside the floodplain, on land not already built in 2023, patches of at least 0.5 ha. Alerts are candidates and must be verified on site.

Population. WorldPop gridded estimate summed over the floodplain part of each unit; an area-level planning estimate only.

Resolution. 10 m pixels: single houses, extra floors and small extensions are not visible.

Candidate from 10 m satellite data. Verify on site before any action. Kinara reports candidate new construction; it does not judge legality or individual households.